

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular & Supplementary Winter Examination-2023
Course: B. Tech. Class: SY Semester : III
Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Code & Name: Electrical Machine- I [BTEEC302]

Max Marks: 60

Date: 04/01/2024

Duration: 03:00 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Explain the principle of operation of single phase transformer and Derive the emf equation of the Transformer	Remember	6
B) Explain the types of Welding Transformer with circuit diagram	Remember	6
C) 40 KVA transformers have iron loss of 450W and full-load copper loss of 850W. If the power factor of the load of 0.8 lagging, calculate: i. Full-load efficiency ii. The load at which maximum efficiency occurs iii. The maximum efficiency.	Create	6
Q.2 Solve Any Two of the following.		12
A) Describe the various three phase transformer connections. Draw circuit diagram and vector diagram of Y- zig zig Y three phase transformer connections	Understand	6
B) Enlist the types of testing of transformer. Explain Routine tests of transformer	Remember	6
C) A 5KVA, 240/2400V, 50Hz single phase transformer has the maximum value of flux density as 1.2 Tesla. If the e.m.f. per turn is 8V. Calculate the number of primary and secondary turns and the primary and secondary current at full load	Create	6
Q.3 Solve Any Two of the following.		12
A) How to determine the magnetic force and torque from energy and co- energy	Understand	6
B) Explain Armature Reaction in DC generator and Effect of Armature Reaction	Evaluate	6

- C) A DC generator has an armature emf of 100v when the useful flux per pole is 20mwb and speed is 800 rpm. Calculate the generated emf Evaluate 6
- a. With the same flux and a speed of 1000 rpm
- b. With flux per pole 24 mwb and speed of 900 rpm

Q.4 Solve Any Two of the following. 12

- A) Write short note on necessity of starter for a DC motor and explain with neat sketch four point starter of a dc shunt motor Remember 6
- B) Derive the expression of the torque in the dc motor. State clearly the meaning of the symbols used Evaluate 6
- C) A 4 pole 250V shunt motor has 500 conductors wave connected. Armature resistance is 0.25 ohm and field resistance is 125 ohm. Flux per pole is 0.02 wb. If motor draw 14A. Current from supply. Find torque and speed of motor. Create 6

Q.5 Solve Any Two of the following. 12

- A) Write a short note on stepper motor. Understand 6
- B) Explain construction and working of reluctance motor with their applications Understand 6
- C) Explain construction and working of BLDC motor with their applications Remember 6

***** End *****